Formal derivation of the EM algorithm for a three-component GMM

Model and notation

We use the same notation as in the lecture slides.

For a three-component Gaussian mixture model,

$$\theta = (\pi_1, \pi_2, \pi_3, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2, \mu_3, \sigma_3^2),$$

with

$$\pi_1 + \pi_2 + \pi_3 = 1.$$

The latent variable is

$$Z_i \in \{1, 2, 3\}.$$

Conditional on the component,

$$X_i \mid Z_i = k \sim N(\mu_k, \sigma_k^2), \qquad k = 1, 2, 3.$$

We write

$$f(x_i; \mu_k, \sigma_k^2)$$

for the Gaussian density with mean μ_k and variance σ_k^2.

Complete likelihood

Following the notation in the slides,

$$P(X, Z \mid \theta) = \prod_{i=1}^{n} \prod_{k=1}^{3} \left[P(Z_i = k \mid \theta) P(X \mid Z_i = k, \theta) \right]^{I(Z_i = k)}.$$

Since

$$P(Z_i = k \mid \theta) = \pi_k$$

and

$$P(X \mid Z_i = k, \theta) = f(x_i; \mu_k, \sigma_k^2),$$

we have

$$P(X, Z \mid \theta) = \prod_{i=1}^{n} \prod_{k=1}^{3} \pi_k^{I(Z_i = k)} f(x_i; \mu_k, \sigma_k^2)^{I(Z_i = k)}.$$

Therefore, the complete log-likelihood is

$$\log(P(X, Z \mid \theta)) = \sum_{i=1}^{n} \sum_{k=1}^{3} I(Z_i = k) \left[\log(\pi_k) + \log\left(f(x_i; \mu_k, \sigma_k^2)\right)\right].$$

For $K = 3$,

$$\begin{aligned}\log(P(X, Z \mid \theta)) = &\sum_{i=1}^{n} I(Z_i = 1) \left[\log(\pi_1) + \log\left(f(x_i; \mu_1, \sigma_1^2)\right)\right] \\ &+ \sum_{i=1}^{n} I(Z_i = 2) \left[\log(\pi_2) + \log\left(f(x_i; \mu_2, \sigma_2^2)\right)\right] \\ &+ \sum_{i=1}^{n} I(Z_i = 3) \left[\log(\pi_3) + \log\left(f(x_i; \mu_3, \sigma_3^2)\right)\right].\end{aligned}$$

E-step

Let

$$\theta^0 = (\pi_1^0, \pi_2^0, \pi_3^0, \mu_1^0, (\sigma_1^2)^0, \mu_2^0, (\sigma_2^2)^0, \mu_3^0, (\sigma_3^2)^0)$$

denote the current value of the parameters.

The E-step computes the posterior distribution

$$P(Z \mid X, \theta^0)$$

and then the expectation of the complete log-likelihood with respect to this posterior:

$$Q(\theta, \theta^0) = E_{Z|X,\theta^0}\left[\log(P(X, Z \mid \theta))\right].$$

Substituting the complete log-likelihood,

$$Q(\theta, \theta^0) = E_{Z|X,\theta^0}\left[\sum_{i=1}^{n} \sum_{k=1}^{3} I(Z_i = k)\Big\{\log(\pi_k) + \log(f(x_i; \mu_k, \sigma_k^2))\Big\}\right].$$

By linearity of expectation,

$$Q(\theta, \theta^0) = \sum_{i=1}^{n} \sum_{k=1}^{3} E_{Z|X,\theta^0}\left[I(Z_i = k)\right]\Big\{\log(\pi_k) + \log(f(x_i; \mu_k, \sigma_k^2))\Big\}.$$

Now,

$$E_{Z|X,\theta^0}\left[I(Z_i = k)\right] = P(Z_i = k \mid X, \theta^0).$$

Therefore,

$$\boxed{Q(\theta, \theta^0) = \sum_{i=1}^{n} \sum_{k=1}^{3} P(Z_i = k \mid X, \theta^0) \left[\log(\pi_k) + \log(f(x_i; \mu_k, \sigma_k^2))\right]}.$$

For $K = 3$,

$$
\begin{aligned}
Q(\theta, \theta^0) = &\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) \left[\log(\pi_1) + \log(f(x_i; \mu_1, \sigma_1^2))\right] \\
&+ \sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0) \left[\log(\pi_2) + \log(f(x_i; \mu_2, \sigma_2^2))\right] \\
&+ \sum_{i=1}^{n} P(Z_i = 3 \mid X, \theta^0) \left[\log(\pi_3) + \log(f(x_i; \mu_3, \sigma_3^2))\right].
\end{aligned}
$$

The probabilities used in the E-step are obtained using Bayes' rule.

For component 1,

$$
\boxed{P(Z_i = 1 \mid X, \theta^0) = \frac{\pi_1^0 f(x_i; \mu_1^0, (\sigma_1^2)^0)}{\pi_1^0 f(x_i; \mu_1^0, (\sigma_1^2)^0) + \pi_2^0 f(x_i; \mu_2^0, (\sigma_2^2)^0) + \pi_3^0 f(x_i; \mu_3^0, (\sigma_3^2)^0)}}.
$$

For component 2,

$$
\boxed{P(Z_i = 2 \mid X, \theta^0) = \frac{\pi_2^0 f(x_i; \mu_2^0, (\sigma_2^2)^0)}{\pi_1^0 f(x_i; \mu_1^0, (\sigma_1^2)^0) + \pi_2^0 f(x_i; \mu_2^0, (\sigma_2^2)^0) + \pi_3^0 f(x_i; \mu_3^0, (\sigma_3^2)^0)}}.
$$

For component 3,

$$
\boxed{P(Z_i = 3 \mid X, \theta^0) = \frac{\pi_3^0 f(x_i; \mu_3^0, (\sigma_3^2)^0)}{\pi_1^0 f(x_i; \mu_1^0, (\sigma_1^2)^0) + \pi_2^0 f(x_i; \mu_2^0, (\sigma_2^2)^0) + \pi_3^0 f(x_i; \mu_3^0, (\sigma_3^2)^0)}}.
$$

These satisfy

$$
P(Z_i = 1 \mid X, \theta^0) + P(Z_i = 2 \mid X, \theta^0) + P(Z_i = 3 \mid X, \theta^0) = 1.
$$

During the M-step, these probabilities are treated as fixed.

M-step

The M-step obtains new parameter estimates

$$
\hat{\theta} = \arg\max_{\theta} Q(\theta, \theta^0).
$$

We now maximise $Q(\theta, \theta^0)$ with respect to each parameter.

Derivation of $\widehat{\mu}_1$

The terms involving μ_1 are

$$
\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) \log(f(x_i; \mu_1, \sigma_1^2)).
$$

For a Gaussian density,

$$\log(f(x_i;\mu_1,\sigma_1^2)) = -\frac{1}{2}\log(2\pi\sigma_1^2) - \frac{(x_i-\mu_1)^2}{2\sigma_1^2}.$$

Therefore,

$$\frac{\partial Q(\theta,\theta^0)}{\partial \mu_1} = \sum_{i=1}^{n} P(Z_i = 1 \mid X,\theta^0)\frac{x_i-\mu_1}{\sigma_1^2}.$$

Set this equal to zero:

$$\sum_{i=1}^{n} P(Z_i = 1 \mid X,\theta^0)(x_i-\mu_1) = 0.$$

Hence,

$$\boxed{\widehat{\mu}_1 = \frac{\sum_{i=1}^{n} P(Z_i = 1 \mid X,\theta^0)x_i}{\sum_{i=1}^{n} P(Z_i = 1 \mid X,\theta^0)}}.$$

By the same argument,

$$\boxed{\widehat{\mu}_2 = \frac{\sum_{i=1}^{n} P(Z_i = 2 \mid X,\theta^0)x_i}{\sum_{i=1}^{n} P(Z_i = 2 \mid X,\theta^0)}}$$

and

$$\boxed{\widehat{\mu}_3 = \frac{\sum_{i=1}^{n} P(Z_i = 3 \mid X,\theta^0)x_i}{\sum_{i=1}^{n} P(Z_i = 3 \mid X,\theta^0)}}.$$

Derivation of $\widehat{\sigma}_1^2$

The terms involving σ_1^2 are

$$-\frac{1}{2}\sum_{i=1}^{n} P(Z_i = 1 \mid X,\theta^0)\left[\log(\sigma_1^2) + \frac{(x_i-\mu_1)^2}{\sigma_1^2}\right].$$

Differentiate with respect to σ_1^2:

$$\frac{\partial Q(\theta,\theta^0)}{\partial \sigma_1^2} = -\frac{1}{2}\sum_{i=1}^{n} P(Z_i = 1 \mid X,\theta^0)\left[\frac{1}{\sigma_1^2} - \frac{(x_i-\mu_1)^2}{(\sigma_1^2)^2}\right].$$

Set this equal to zero:

$$\sum_{i=1}^{n} P(Z_i = 1 \mid X,\theta^0)\left[\frac{1}{\sigma_1^2} - \frac{(x_i-\mu_1)^2}{(\sigma_1^2)^2}\right] = 0.$$

Multiplying by $(\sigma_1^2)^2$ gives

$$\sigma_1^2 \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0) = \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)(x_i - \mu_1)^2.$$

Therefore,

$$\boxed{\widehat{\sigma}_1^2 = \frac{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)(x_i - \widehat{\mu}_1)^2}{\sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)}}.$$

Similarly,

$$\boxed{\widehat{\sigma}_2^2 = \frac{\sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0)(x_i - \widehat{\mu}_2)^2}{\sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0)}}$$

and

$$\boxed{\widehat{\sigma}_3^2 = \frac{\sum_{i=1}^{n} P(Z_i = 3 \mid X, \theta^0)(x_i - \widehat{\mu}_3)^2}{\sum_{i=1}^{n} P(Z_i = 3 \mid X, \theta^0)}}.$$

Derivation of $\widehat{\pi}_1$, $\widehat{\pi}_2$ and $\widehat{\pi}_3$

The mixing proportions satisfy the constraint

$$\pi_1 + \pi_2 + \pi_3 = 1.$$

The part of $Q(\theta, \theta^0)$ involving the mixing proportions is

$$Q_\pi = \sum_{i=1}^{n} \sum_{k=1}^{3} P(Z_i = k \mid X, \theta^0) \log(\pi_k).$$

To maximise this subject to

$$\pi_1 + \pi_2 + \pi_3 = 1,$$

we introduce the Lagrange multiplier λ:

$$\mathcal{L} = \sum_{i=1}^{n} \sum_{k=1}^{3} P(Z_i = k \mid X, \theta^0) \log(\pi_k) + \lambda \left(1 - \pi_1 - \pi_2 - \pi_3\right).$$

For $k = 1, 2, 3$,

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{\sum_{i=1}^{n} P(Z_i = k \mid X, \theta^0)}{\pi_k} - \lambda.$$

Setting this equal to zero gives

$$\pi_k = \frac{\sum_{i=1}^{n} P(Z_i = k \mid X, \theta^0)}{\lambda}.$$

Summing over $k = 1, 2, 3$,

$$\sum_{k=1}^{3} \pi_k = \frac{1}{\lambda} \sum_{i=1}^{n} \sum_{k=1}^{3} P(Z_i = k \mid X, \theta^0).$$

Since

$$\sum_{k=1}^{3} \pi_k = 1$$

and

$$\sum_{k=1}^{3} P(Z_i = k \mid X, \theta^0) = 1,$$

we have

$$1 = \frac{1}{\lambda} \sum_{i=1}^{n} 1 = \frac{n}{\lambda}.$$

Therefore,

$$\lambda = n.$$

Hence,

$$\boxed{\widehat{\pi}_1 = \frac{1}{n} \sum_{i=1}^{n} P(Z_i = 1 \mid X, \theta^0)}$$

$$\boxed{\widehat{\pi}_2 = \frac{1}{n} \sum_{i=1}^{n} P(Z_i = 2 \mid X, \theta^0)}$$

and

$$\boxed{\widehat{\pi}_3 = \frac{1}{n} \sum_{i=1}^{n} P(Z_i = 3 \mid X, \theta^0)}.$$

Notice that

$$\widehat{\pi}_1 + \widehat{\pi}_2 + \widehat{\pi}_3 = 1.$$

Final EM updates for $K = 3$

Starting from the current parameter values θ^0:

E-step

For each observation i, compute

$$P(Z_i = 1 \mid X, \theta^0) = \frac{\pi_1^0 f(x_i; \mu_1^0, (\sigma_1^2)^0)}{\sum_{j=1}^3 \pi_j^0 f(x_i; \mu_j^0, (\sigma_j^2)^0)},$$

$$P(Z_i = 2 \mid X, \theta^0) = \frac{\pi_2^0 f(x_i; \mu_2^0, (\sigma_2^2)^0)}{\sum_{j=1}^3 \pi_j^0 f(x_i; \mu_j^0, (\sigma_j^2)^0)},$$

and

$$P(Z_i = 3 \mid X, \theta^0) = \frac{\pi_3^0 f(x_i; \mu_3^0, (\sigma_3^2)^0)}{\sum_{j=1}^3 \pi_j^0 f(x_i; \mu_j^0, (\sigma_j^2)^0)}.$$

M-step

With these probabilities fixed,

$$\boxed{\widehat{\pi}_k = \frac{1}{n} \sum_{i=1}^n P(Z_i = k \mid X, \theta^0), \qquad k = 1, 2, 3}$$

$$\boxed{\widehat{\mu}_k = \frac{\sum_{i=1}^n P(Z_i = k \mid X, \theta^0) x_i}{\sum_{i=1}^n P(Z_i = k \mid X, \theta^0)}, \qquad k = 1, 2, 3}$$

and

$$\boxed{\widehat{\sigma}_k^2 = \frac{\sum_{i=1}^n P(Z_i = k \mid X, \theta^0)(x_i - \widehat{\mu}_k)^2}{\sum_{i=1}^n P(Z_i = k \mid X, \theta^0)}, \qquad k = 1, 2, 3}.$$

The new estimate is therefore

$$\hat{\theta} = (\widehat{\pi}_1, \widehat{\pi}_2, \widehat{\pi}_3, \widehat{\mu}_1, \widehat{\sigma}_1^2, \widehat{\mu}_2, \widehat{\sigma}_2^2, \widehat{\mu}_3, \widehat{\sigma}_3^2).$$

We then replace the current parameter values by the new estimates,

$$\theta^0 \leftarrow \hat{\theta},$$

and repeat the E-step and M-step until the incomplete log-likelihood changes by less than a small value ϵ.